Codomax

Digital Solutions & Beyond

Cloud Engineering Internship Report: AWS Administration & EC2 Deployment

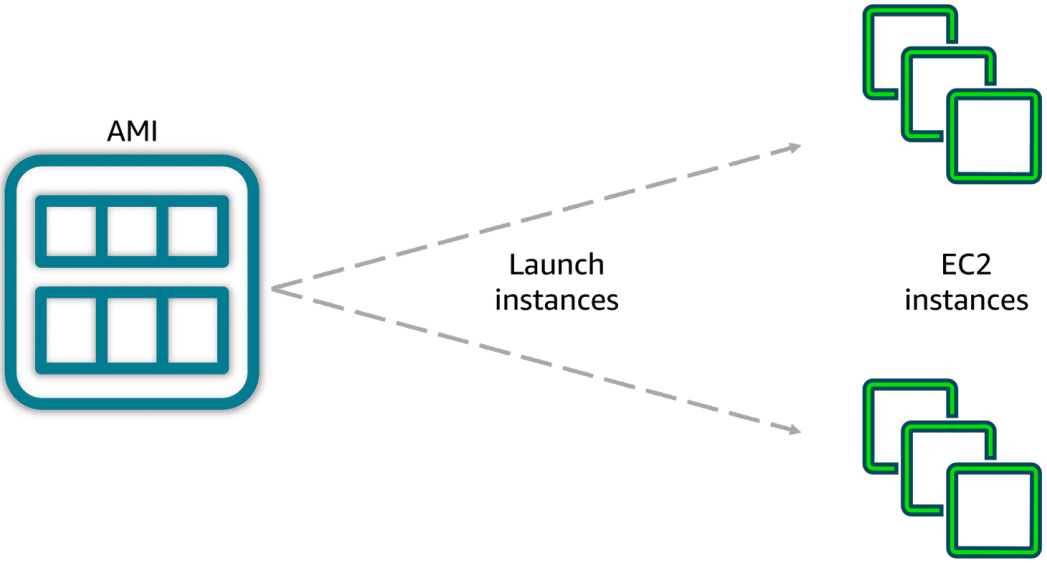
Author: Fahad | Module 2 Comprehensive Task Breakdown

Executive Summary

This report documents the practical tasks completed during Module 2 of the cloud engineering internship. The focus spans core AWS Identity and Access Management (IAM) configurations, implementation of foundational security principles, launching and connecting to Amazon EC2 instances, configuring network security groups, and deploying a functional web server displaying a custom HTML website.

Overview Architecture & Concepts

The diagram below illustrates the relationship between Amazon Machine Images (AMIs) and launching virtual server instances (EC2) in the AWS Cloud architecture environment.

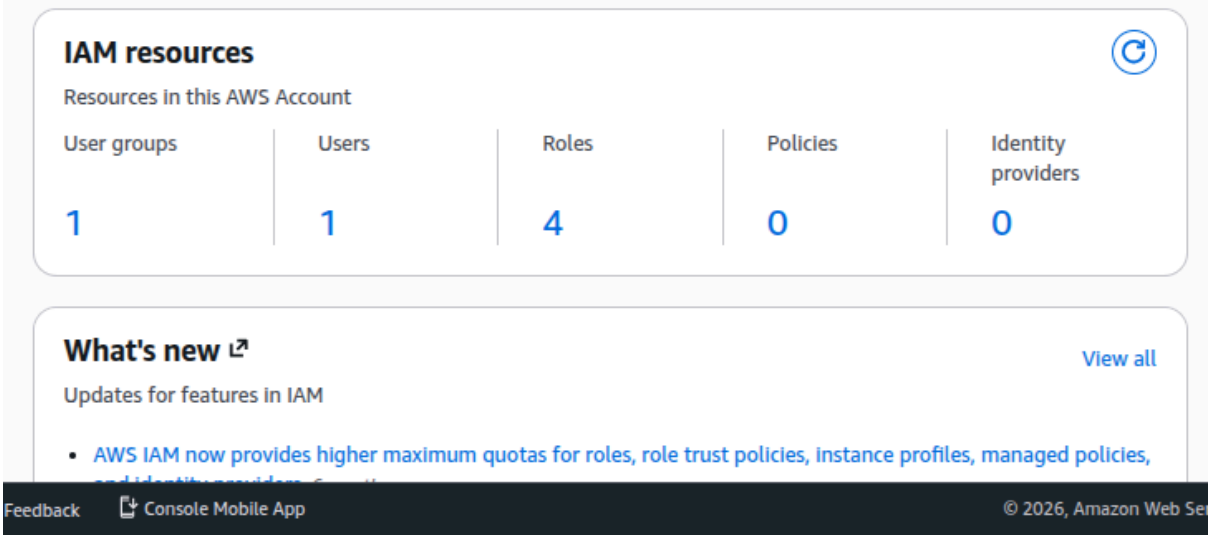


Task 1: IAM Users, Groups, Roles, Policies, and Permissions

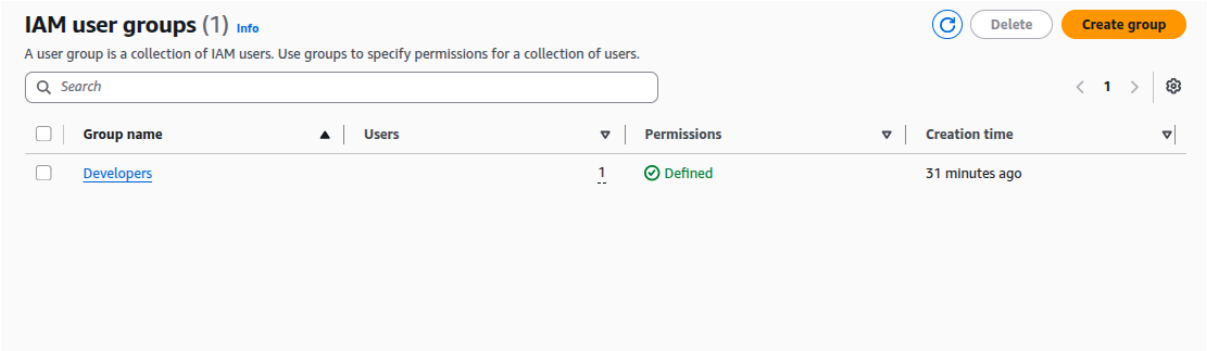
In Task 1, Identity and Access Management (IAM) structures were established to manage identities and access control securely within the AWS account. The implementation included creating an IAM user group (Developers), provisioning individual IAM users (dev-user1), creating IAM roles (EC2-S3-ReadOnly-Role), and attaching policies to enforce permission boundaries.

Task 1 Artifacts

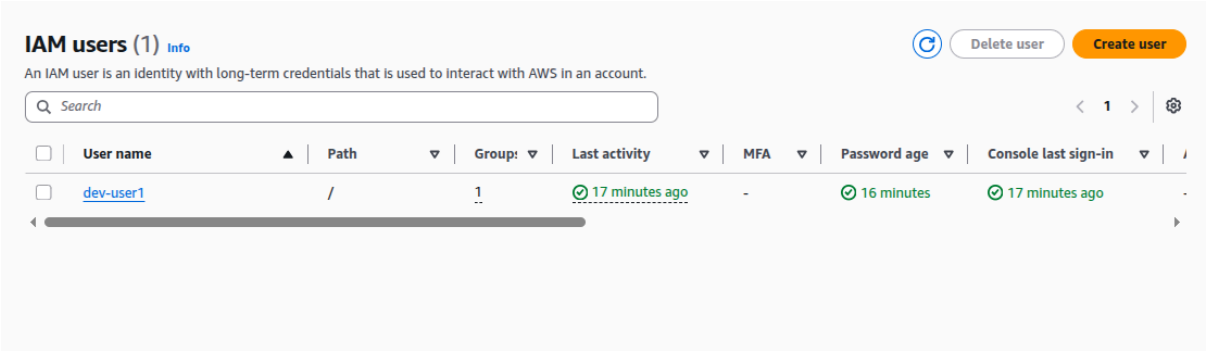
IAM Resources Overview & Configuration:



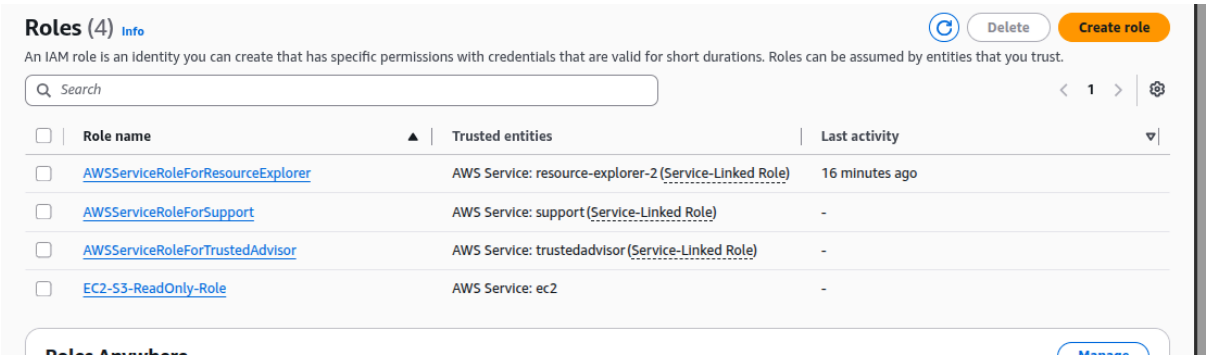
Configured IAM User Groups:



Provisioned IAM Users:



Defined IAM Roles:

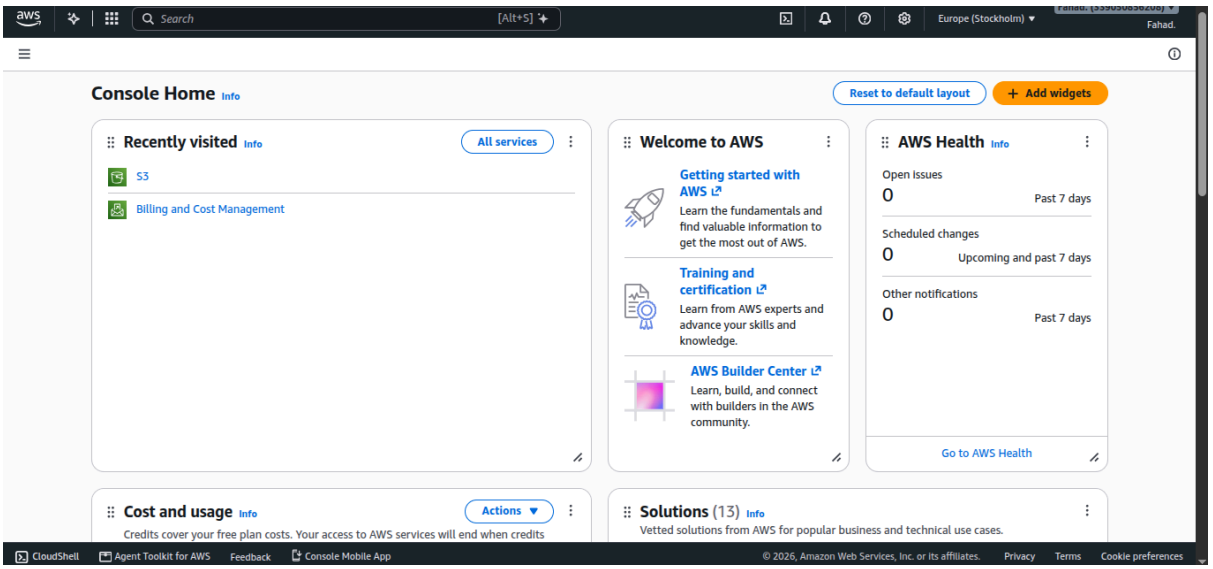


Task 2: Basic AWS Security & Principle of Least Privilege

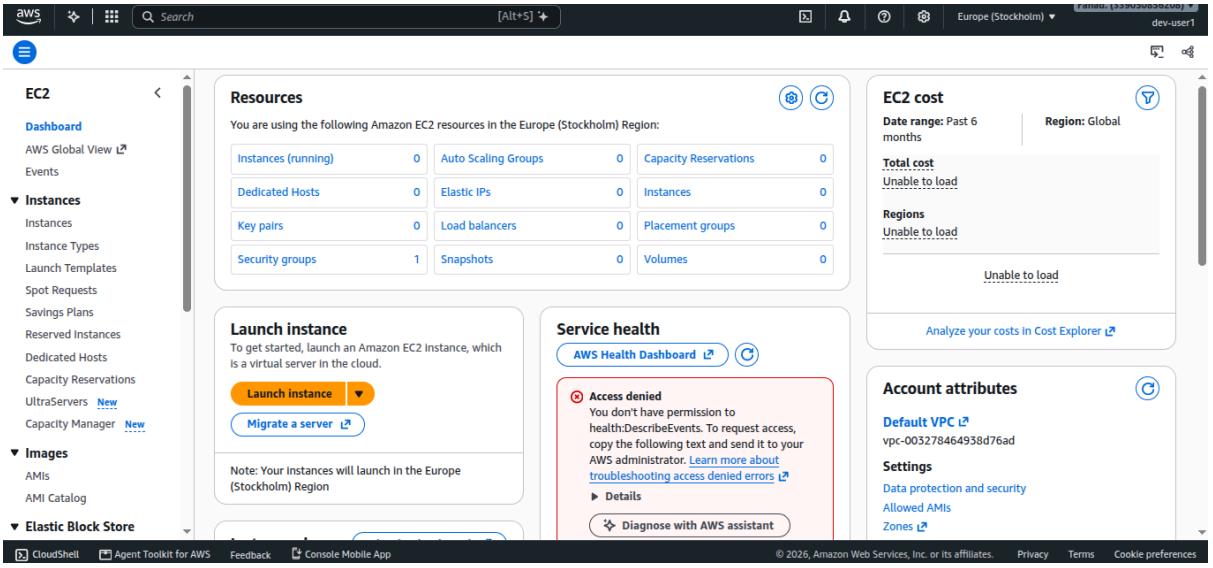
To maintain robust security posture, access permissions were restricted using the Principle of Least Privilege. Users were assigned only the exact permissions needed for their defined roles, avoiding root account usage for daily tasks. Access restriction tests confirmed that unauthorized service calls (such as health status events) are explicitly denied by default policy boundaries.

Task 2 Artifacts

AWS Console View & Access Control Verification:



Access Denied Verification under Least Privilege Policies:

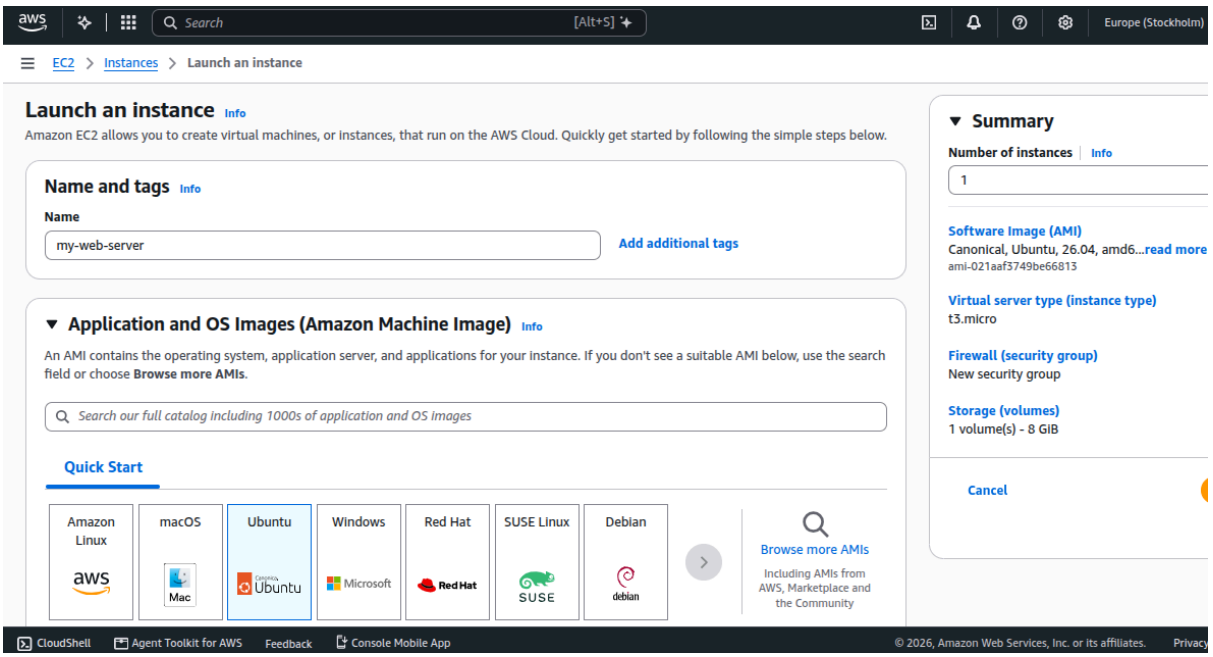


Task 3: Launch and Configure an Amazon EC2 Instance

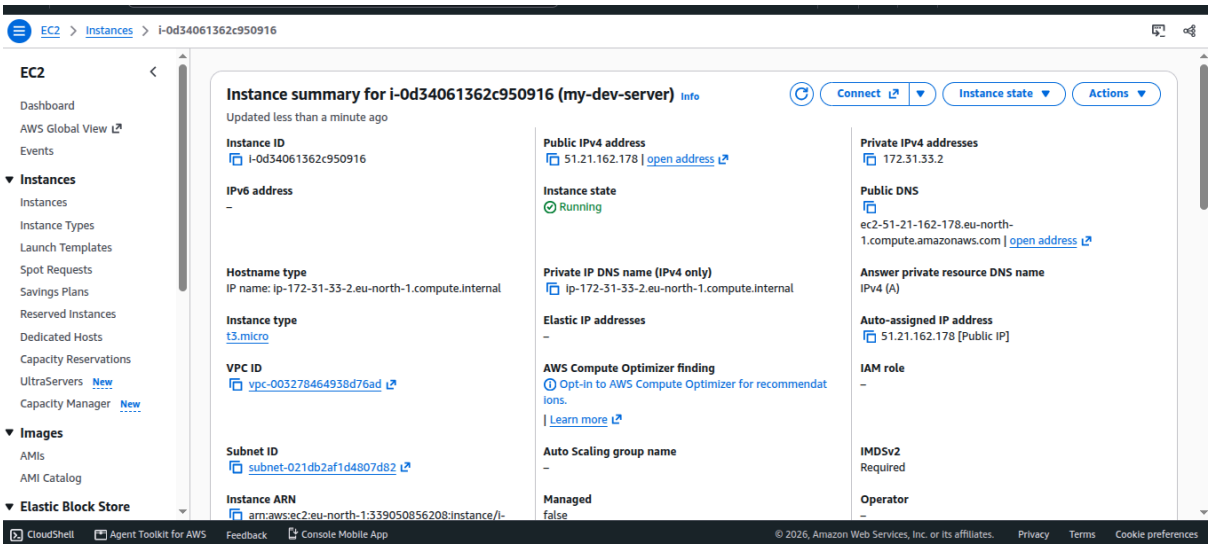
An Amazon EC2 instance named "my-web-server" (instance ID: i-0d34061362c950916) was configured and launched within the Europe (Stockholm) region. The setup utilized Ubuntu Server 26.04 LTS on a t3.micro instance type within the default VPC framework.

Task 3 Artifacts

Launch Instance Wizard Configuration:



EC2 Instance Summary & Status:

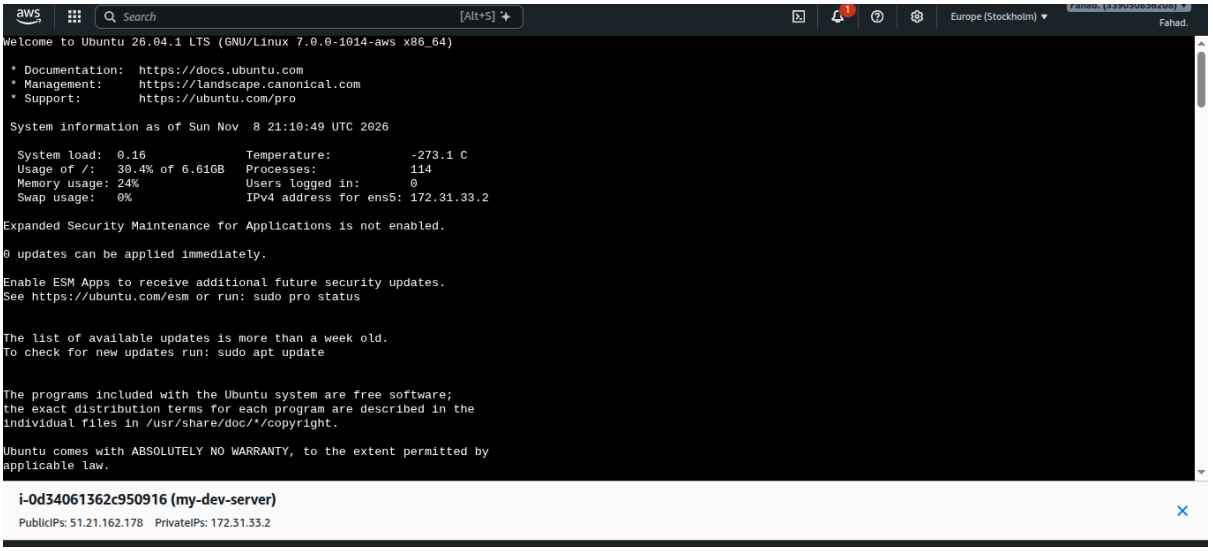


Task 4: Configure a Security Group and Connect to EC2

Network access controls were configured via EC2 Security Groups to open necessary ports (Port 22 for SSH/Console management and Port 80 for HTTP web traffic). Connection to the virtual server instance was established successfully via the AWS web-based SSH terminal.

Task 4 Artifacts

Connected Linux Terminal Shell Session (my-web-server):



Task 5: Install a Web Server and Deploy a Basic HTML Website

An Apache web server HTTP daemon was installed on the Ubuntu EC2 instance. A custom HTML page was authored and hosted on the server, publicly reachable via the instance's elastic Public IPv4 address (51.21.162.178).

Task 5 Artifacts

Publicly Accessible Deployed HTML Website:

